

Greenbone vs Nessus

Vulnerability Scan Comparison Report

Objective

The objective of this task is to perform a vulnerability scan on the same Windows 10 VM using Greenbone Community Edition (OpenVAS) and compare the results with the previously conducted Nessus Essentials scan to analyze the differences in detection capabilities between both tools.

Environment Setup

For performing this comparison task, the following tools and environment were configured:

Scanner 1 — Nessus Essentials

Tool	Nessus Essentials by Tenable
Installed on	Host Machine (Windows)
Access URL	https://localhost:8834
Scan Type	Basic Network Scan

Scanner 2 — Greenbone Community Edition

Tool	Greenbone Community Edition (OpenVAS)
Deployment	Docker Desktop on Host Machine (Windows)
Docker Compose	Used to deploy all 20+ containers
Access URL	http://127.0.0.1:9392
Scan Type	Full and Fast

Target Machine

OS	Windows 10
Platform	Oracle VirtualBox
IP Address	192.168.56.101
Network Mode	Host-only Adapter

Basic Understanding:

What is Greenbone / OpenVAS?

Greenbone Community Edition is a free and open-source vulnerability scanner built on top of the OpenVAS engine. It is developed and maintained by Greenbone Networks and is widely used in the cybersecurity community as an alternative to commercial scanners like Nessus.

Major functions of Greenbone include:

- Network Vulnerability Testing (NVT) with 200,000+ checks.
- CVE-based vulnerability detection and reporting.
- Web-based dashboard for scan management and results.

Nessus vs Greenbone — Key Differences

Feature	Nessus	Greenbone
Developer	Tenable	Greenbone Networks
Type	Commercial (Free Essentials)	Open Source
Engine	Nessus Engine	OpenVAS Engine
Plugin Database	Tenable Plugins	NVT (Community Feed)
Interface	Web UI (port 8834)	Web UI (port 9392)
Deployment	Direct Installation	Docker Containers

Challenges Faced During Setup

Several challenges were encountered during the setup of Greenbone that are worth documenting:

Challenge	Resolution
rsync port 873 blocked in Pakistan	Switched to Docker which uses HTTP instead of rsync
ProtonVPN website blocked	Used Docker-based deployment to bypass region restrictions(To mitigate the issue of rsync protocol VPN was tried but this approach also get failed)
scap-data container unhealthy	Force-recreated container using docker compose command(and scap data feed was dropped because its health check got failed and greenbone in browser)

	Since all 20/21 containers has been start up— scap-data ka data had already copied(visible when checked in logs and only the health check was failed).
VM not reachable from Docker	Disabled Windows Firewall on VM temporarily for scan

Due to regional network restrictions in Pakistan, the rsync protocol (port 873) used by Greenbone for feed synchronization was blocked at the ISP level. This prevented the vulnerability feeds (NVT, SCAP, CERT, and GVM_DATA) from syncing when Greenbone was installed directly on Kali Linux. To resolve this issue, Greenbone Community Edition was deployed using Docker Desktop on the host Windows machine. Docker-based deployment communicates over HTTP(port 80) instead of rsync, which successfully bypassed the regional restriction. Once all four feeds completed synchronization, the scan task became available and the vulnerability scan was launched.

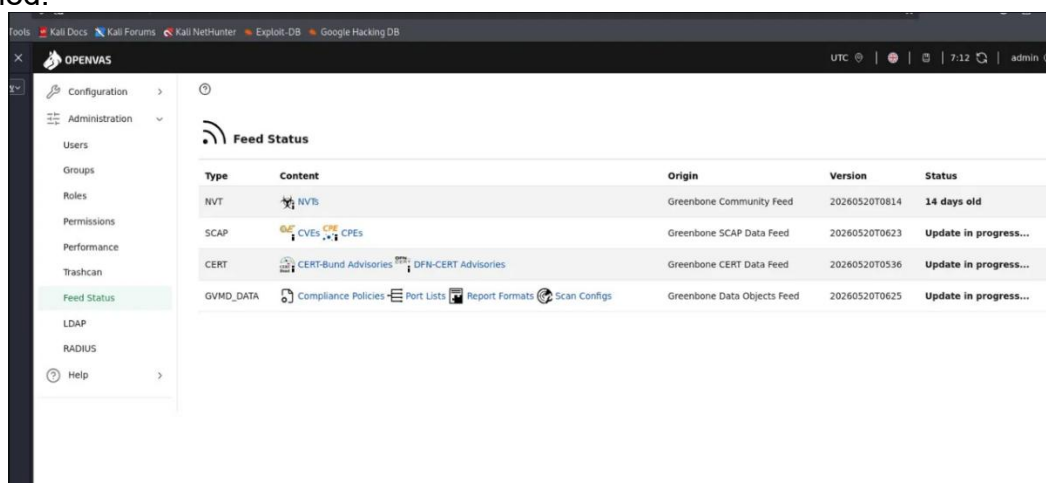


Figure 1: Feeds synchronization issue

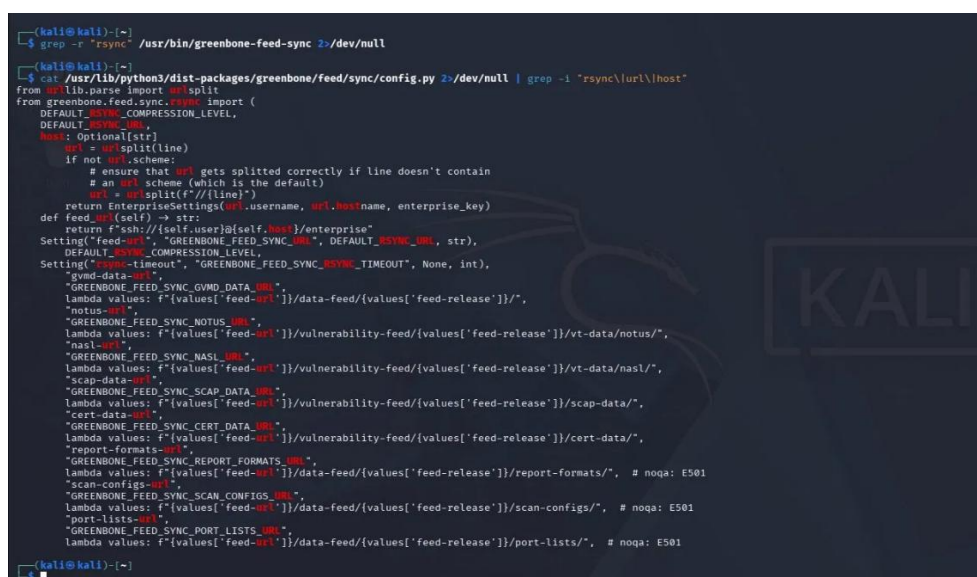


Figure 2: Rsync protocol issue

```

Administrator: Windows PowerShell
✓Image registry.community.greenbone.net/community/gvm-config:latest Pulled 1224.98
✓Image registry.community.greenbone.net/community/gvmd:stable Pulled 2830.78
✓Image registry.community.greenbone.net/community/gvm-tools Pulled 1276.36
✓Image registry.community.greenbone.net/community/ospd-openvas:stable Pulled 1684.25
✓Image registry.community.greenbone.net/community/vulnerability-tests Pulled 1973.05
✓Image registry.community.greenbone.net/community/redis-server Pulled 1219.75
✓Image registry.community.greenbone.net/community/pg-gvm-migrator:stable Pulled 2351.35
✓Image registry.community.greenbone.net/community/pg-gvm:stable Pulled 2257.05
✓Image registry.community.greenbone.net/community/nginx:latest Pulled 1315.53
✓Image registry.community.greenbone.net/community/dfn-cert-data Pulled 377.35
✓Image registry.community.greenbone.net/community/openvas-scanner:stable Pulled 1598.15
✓Image registry.community.greenbone.net/community/gsad:stable Pulled 1226.45
✓Image registry.community.greenbone.net/community/cert-bund-data Pulled 465.45
✓Image registry.community.greenbone.net/community/scap-data Pulled 6795.65
✓Image registry.community.greenbone.net/community/notus-data Pulled 1317.95
✓Network greenbone-community-edition_default Created 0.50
✓Volume greenbone-community-edition_vt_data_vol Created 0.25
✓Volume greenbone-community-edition_nginx_certificates_vol Created 0.00
✓Volume greenbone-community-edition_psql_data_vol Created 0.00
✓Volume greenbone-community-edition_notus_data_vol Created 0.15
✓Volume greenbone-community-edition_ospd_openvas_socket_vol Created 0.00
✓Volume greenbone-community-edition_gsa_data_vol Created 0.00
✓Volume greenbone-community-edition_openvas_data_vol Created 0.15
✓Volume greenbone-community-edition_scap_data_vol Created 0.00
✓Volume greenbone-community-edition_redis_socket_vol Created 0.15
✓Volume greenbone-community-edition_cert_data_vol Created 0.00
✓Volume greenbone-community-edition_psql_socket_vol Created 0.15
✓Volume greenbone-community-edition_nginx_config_vol Created 0.25
✓Volume greenbone-community-edition_openvas_log_data_vol Created 0.00
✓Volume greenbone-community-edition_data_objects_vol Created 0.00
✓Volume greenbone-community-edition_gvmd_data_vol Created 0.15
✓Volume greenbone-community-edition_gpg_data_vol Created 0.00
✓Volume greenbone-community-edition_gvmd_socket_vol Created 0.00
✗Container greenbone-community-edition-scap-data-1 Error dependency scap-data failed to start 100.75
✓Container greenbone-community-edition-gsa-1 Started 23.05
✓Container greenbone-community-edition-cert-bund-data-1 Healthy 65.25
✓Container greenbone-community-edition-notus-data-1 Healthy 70.15
... 17 more
dependency failed to start: container greenbone-community-edition-scap-data-1 is unhealthy

```

Figure 3: Scap data synchronization issue

```

PS D:\greenbone> docker compose up -d --no-deps gvmd gsad nginx ospd-openvas openvasd
[+] up 5/5
✓Container greenbone-community-edition-ospd-openvas-1 Running
✓Container greenbone-community-edition-openvasd-1 Running
✓Container greenbone-community-edition-gvmd-1 Started
✓Container greenbone-community-edition-gsad-1 Started
✓Container greenbone-community-edition-nginx-1 Started

What's next:
  Filter, search, and stream logs from all your Compose services
  in one place with Docker Desktop's Logs view. docker-desktop://dashboard/logs?appId=greenbone-community-edition
PS D:\greenbone>

```

Figure 4: all 5 containers started(scap data feed drop)

What is Dockers and How does it resolve feed synchronization issue?

Docker is a containerization platform that allows applications to run inside isolated environments called containers. Unlike virtual machines, containers are lightweight, fast to start, and share the host operating system — making them efficient for deploying complex software like Greenbone Community Edition.

The difference in approach which is used by Dockers is:

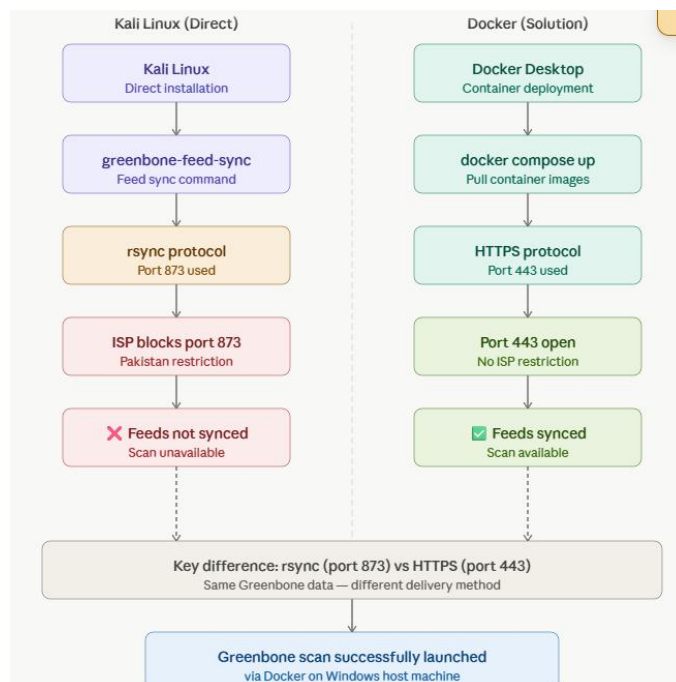


Figure 5: Distinguishing approach flowchart

In simple terms, the issue was not resolved by changing a setting but by switching to an entirely different deployment method that used an open and unrestricted communication channel. Once all containers were running and feeds completed synchronization, the Greenbone web interface became fully functional and the vulnerability scan was successfully launched.

Scan Methodology

Setting Up Greenbone via Docker

The following steps were taken to deploy Greenbone Community Edition using Docker Desktop on the host Windows machine:

1. Docker Desktop was installed on the host Windows machine.
2. WSL 2 (Windows Subsystem for Linux) was updated to support Docker containers.
3. A docker-compose.yml file was downloaded from Greenbone's official documentation.
4. All 20+ Greenbone containers were pulled and started using 'docker compose up -d' command.
5. Feed synchronization was completed for NVT, SCAP, CERT, and GVM_DATA.
6. Greenbone web interface was accessed via <http://127.0.0.1:9392>.

Launching the Scan

After successful setup, the scan was configured and launched with the following settings:

Scan Name	VM Scan
Target IP	192.168.56.101
Scan Config	Full and Fast
Alive Test	Consider Hosts as Alive
Port List	All IANA assigned TCP

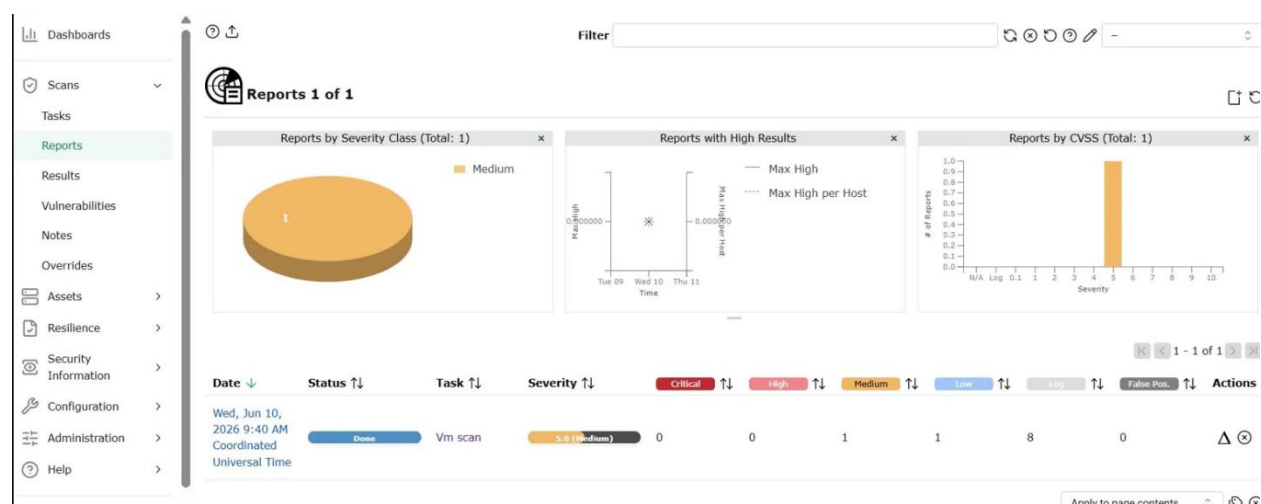
Greenbone Scan Results

The scan was completed successfully and the following results were obtained:

Results Summary

Severity	Count
Critical	0
High	0
Medium	1
Low	1
Log / Info	8
Total	10

The following screenshot validate the results:



Detailed Findings

Vulnerability	Severity	Port	Host
DCE/RPC and MSRPC Services Enumeration	Log	135/tcp	192.168.56.101
SMB/CIFS Server Detection	Log	445/tcp	192.168.56.101
SMB/CIFS Server Detection	Log	139/tcp	192.168.56.101
OS Detection Consolidation and Reporting	Log	general/tcp	192.168.56.101
Traceroute	Log	general/tcp	192.168.56.101
SMB Remote Version Detection	Log	445/tcp	192.168.56.101
Hostname Determination Reporting	Log	general/tcp	192.168.56.101
CPE Inventory	Log	general/CPE-T	192.168.56.101



Vulnerability ↓	Severity ↑	QoD ↑	Host IP ↑	Name ↑	Location ↑	EPSS Score ↑	Percentile ↑	Created ↑
DCE/RPC and MSRPC Services Enumeration	0.0 (Log)	80 %	192.168.56.101		135/tcp	N/A	N/A	Wed, Jun 10, 2026 9:45 AM Coordinated Universal Time
SMB/CIFS Server Detection	0.0 (Log)	80 %	192.168.56.101		445/tcp	N/A	N/A	Wed, Jun 10, 2026 9:45 AM Coordinated Universal Time
SMB/CIFS Server Detection	0.0 (Log)	80 %	192.168.56.101		139/tcp	N/A	N/A	Wed, Jun 10, 2026 9:45 AM Coordinated Universal Time
OS Detection Consolidation and Reporting	0.0 (Log)	80 %	192.168.56.101		general/tcp	N/A	N/A	Wed, Jun 10, 2026 9:45 AM Coordinated Universal Time
Traceroute	0.0 (Log)	80 %	192.168.56.101		general/tcp	N/A	N/A	Wed, Jun 10, 2026 9:46 AM Coordinated Universal Time
SMB Remote Version Detection	0.0 (Log)	80 %	192.168.56.101		445/tcp	N/A	N/A	Wed, Jun 10, 2026 9:46 AM Coordinated Universal Time
Hostname Determination Reporting	0.0 (Log)	80 %	192.168.56.101		general/tcp	N/A	N/A	Wed, Jun 10, 2026 9:56 AM Coordinated Universal Time
CPE Inventory	0.0 (Log)	80 %	192.168.56.101		general/CPE-T	N/A	N/A	Wed, Jun 10, 2026 9:56 AM Coordinated Universal Time

Nessus vs Greenbone — Comparison

Both scanners were run against the same target (Windows 10 VM, IP: 192.168.56.101). The comparison of results is presented below:

Results Comparison Table

Category	Nessus Essentials	Greenbone (OpenVAS)
Total Findings	16	10
Critical	0	0
High	0	0
Medium	1	1
Low	1	1
Info / Log	14	8
SMB Signing Detected	Yes (before fix)	No (after fix applied)
Scan Type	Basic Network Scan	Full and Fast
Scan Duration	~8 minutes	~45 minutes

Key Observations

The following key observations were made from the comparison of both scanners:

7. Nessus detected 16 vulnerabilities while Greenbone detected 10 — a difference of 6 findings.
8. Nessus detected the 'SMB Signing not required' vulnerability (Medium, CVSS 5.3) which was then fixed before the Greenbone scan was run — hence Greenbone did not flag this finding.
9. Both scanners detected SMB-related services on port 445 and 139 but differed in depth of analysis.
10. Greenbone's Full and Fast scan took significantly longer (~45 minutes) compared to Nessus Basic Scan (~8 minutes).
11. Both tools use different plugin/NVT databases which accounts for differences in findings count.

The screenshots given below shows the details of vulnerabilities regarding SMB which clearly shows that it doesnot flag any “SMB signing requirement”.

SMB Remote Version Detection

0.0 (Log)80 %192.168.56.101445/tcpN/AN/A

Wed, Jun 10, 2026 9:46 AM Coordinated Universal Time

Summary

Detection of Server Message Block(SMB).

This script sends SMB Negotiation request and try to get the version from the response.

Detection Result

SMBv2 and SMBv3 are enabled on remote target

Detection Method

Details: SMB Remote Version Detection OID: 1.3.6.1.4.1.25623.1.0.807830

Version used: 2023-07-26T05:05:09Z

SMB/CIFS Server Detection	6.6 (Log)	80 %	192.168.56.101	445/tcp	N/A	N/A	Wed, Jun 10, 2026 9:45 AM Coordinated Universal Time
Summary This script detects whether port 445 and 139 are open and if they are running a CIFS/SMB server. Detection Result A CIFS server is running on this port. Detection Method Details: SMB/CIFS Server Detection OED: 1.3.6.1.4.1.25623.1.0.11011 Version used: 2023-08-01T13:29:10Z							
SMB/CIFS Server Detection	6.6 (Log)	80 %	192.168.56.101	139/tcp	N/A	N/A	Wed, Jun 10, 2026 9:45 AM Coordinated Universal Time
Summary This script detects whether port 445 and 139 are open and if they are running a CIFS/SMB server. Detection Result A SMB server is running on this port. Detection Method Details: SMB/CIFS Server Detection OED: 1.3.6.1.4.1.25623.1.0.11011 Version used: 2023-08-01T13:29:10Z							

Why Results Differ Between Scanners

Reason	Explanation
Different Plugin Databases	Nessus uses Tenable plugins; Greenbone uses NVT community feed — each has unique checks
SMB Signing already fixed	The Medium vulnerability was fixed before Greenbone scan, so it was not detected
Scan Configuration	Nessus used Basic scan; Greenbone used Full and Fast — different depth of checks
Detection Algorithms	Each scanner uses different methods to probe and analyze vulnerabilities

Nessus vulnerability scan results (screenshots) are also attach here as a reference:

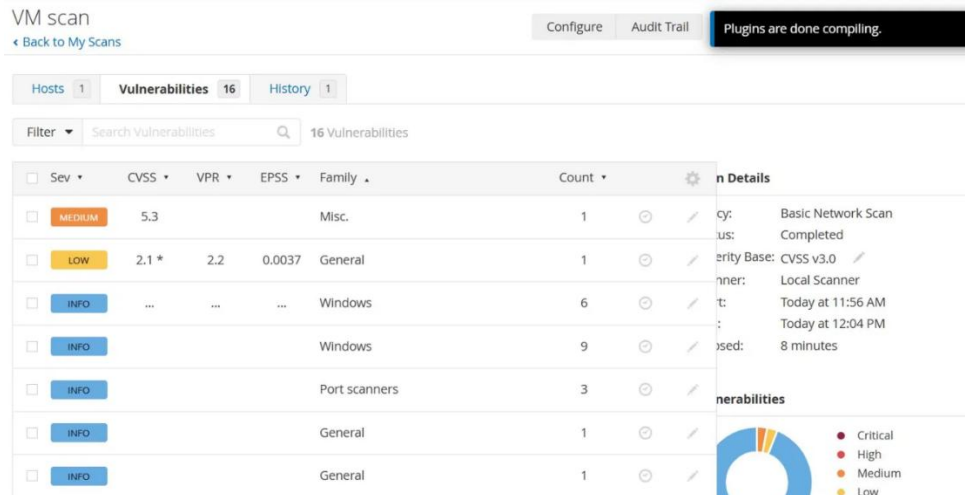


Figure 6: Initial scanning results

☐	INFO	General	1		
☐	INFO	General	1		
☐	INFO	Misc.	1		
☐	INFO	General	1		
☐	INFO	General	1		
☐	INFO	Settings	1		
☐	INFO	General	1		
☐	INFO	General	1		
☐	INFO	Settings	1		
☐	INFO	Settings	1		
☐	INFO	General	1		

Plugins are done compiling.

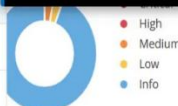


Figure 7: Initial Scanning results

tenable

Nessus Essentials

Scans

Settings

FOLDERS

My Scans

All Scans

Trash

RESOURCES

Policies

Plugin Rules

Tenable News

Frequently asked

MEDIUM

SMB Signing not required

Description

Signing is not required on the remote SMB server. An unauthenticated, remote attacker can exploit this to conduct man-in-the-middle attacks against the SMB server.

Solution

Enforce message signing in the host's configuration. On Windows, this is found in the policy setting 'Microsoft network server: Digitally sign communications (always)'. On Samba, the setting is called 'server signing'. See the 'see also' links for further details.

See Also

<http://www.nessus.org/u?df39b8b3>
<http://technet.microsoft.com/en-us/library/cc731957.aspx>
<http://www.nessus.org/u?74b80723>
<https://www.samba.org/samba/docs/current/man-html/smb.conf.5.html>
<http://www.nessus.org/u?a3cac4ea>

Output

No output recorded.

To see debug logs, please visit individual host

Port	Hosts
445 / tcp / cifs	192.168.56.101

CVSS v3.0 Temporal Score: 4.6

CVSS v2.0 Base Score: 5.0

CVSS v2.0 Temporal Score: 3.7

CVSS v2.0 Vector:

CVSS2#AV:N/AC:L/Au:N/C:N/I:P/A:N

CVSS v2.0 Temporal Vector:

CVSS2#E:U/RL:OF/RC:C

Vulnerability Information

Exploit Available: true

Exploit Ease: Exploits are available

Vulnerability Pub Date: January 17, 2012

Figure 8: Vulnerability detail(medium)

Conclusion

This task successfully demonstrated the use of two industry-standard vulnerability scanners — Nessus Essentials and Greenbone Community Edition — to scan the same Windows 10 VM and compare their results.

Nessus detected 16 vulnerabilities including the critical 'SMB Signing not required' finding (CVSS 5.3) which was subsequently fixed. Greenbone, running after the fix was applied, detected 10 findings and did not flag the SMB Signing vulnerability — confirming that the fix was effective.

The comparison highlights that no single vulnerability scanner is complete on its own. Using multiple tools provides broader coverage and helps ensure that security professionals do not miss critical vulnerabilities. Each tool brings unique detection capabilities, plugin databases, and scanning methodologies that complement each other in a real-world security assessment.